

Prob 1.

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$$\|x\|_\infty = \max_i |x_i| \leq \sum_{i=1}^n |x_i| = \|x\|_1$$

$$\|x\|_1 = \sum_{i=1}^n |x_i| \leq \sum_{i=1}^n \max_i |x_i| = n \|x\|_\infty$$

$$\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2} \leq \sqrt{\sum_{i=1}^n \max_i x_i^2} = \sqrt{\sum_{i=1}^n (\max_i |x_i|)^2} = \sqrt{n \|x\|_\infty^2} = \sqrt{n} \|x\|_\infty$$

$$\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2} \geq \sqrt{\max_i |x_i|^2} = \sqrt{(\max_i |x_i|)^2} = \|x\|_\infty$$

$$\|x\|_2^2 = \sum_{i=1}^n x_i^2 \leq \max_i |x_i| \sum_{i=1}^n |x_i| = \|x\|_\infty \|x\|_1$$

Since $\|x\|_\infty \leq \|x\|_2$, we have

$$\|x\|_2 \leq \|x\|_1$$

for $x \neq 0$. Clearly, the inequality is valid also at $x = 0$. Finally, using the inequality

$$2ab \leq a^2 + b^2$$

it can be shown by induction that $\|x\|_1 \leq \sqrt{n} \|x\|_2$. For suppose

$$\left(\sum_{i=1}^{n-1} |x_i| \right)^2 \leq (n-1) \sum_{i=1}^{n-1} x_i^2$$

then

$$\begin{aligned} \left(\sum_{i=1}^n |x_i| \right)^2 &= \left(\sum_{i=1}^{n-1} |x_i| \right)^2 + 2|x_n| \left(\sum_{i=1}^{n-1} |x_i| \right) + |x_n|^2 \\ &\leq (n-1) \sum_{i=1}^{n-1} x_i^2 + \sum_{i=1}^{n-1} (|x_n|^2 + |x_i|^2) + |x_n|^2 \\ &= n \sum_{i=1}^n x_i^2 \end{aligned}$$

$$\begin{aligned} (a-b)^2 &\geq 0 \\ a^2 - 2ab + b^2 &\geq 0 \\ \therefore a^2 + b^2 &\geq 2ab \\ 2ab &\leq a^2 + b^2 \leq 2(a^2 + b^2) \end{aligned}$$

Working correctly the case $n=2$ is worth 2 points.

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Alternate way of showing that

$$\forall x \in \mathbb{R}^n, \quad \|x\|_1 \leq \sqrt{n} \|x\|_2$$

Suppose that $\forall x \in \mathbb{R}^n, \|x\|_1 \leq K \|x\|_2$.

We seek to find the smallest K .

$$\forall x \in \mathbb{R}^n, \quad K \geq \frac{\|x\|_1}{\|x\|_2} \\ x \neq 0$$

$$\therefore K \geq \max_{\substack{x \in \mathbb{R}^n \\ x \neq 0}} \frac{\|x\|_1}{\|x\|_2} = \max_{\|x\|_2=1} \sum_{i=1}^n |x_i|$$

Since for $x = (x_1, \dots, x_n) \in \mathbb{R}^n$,

$$\|(|x_1|, \dots, |x_n|)\|_\alpha = \| (x_1, \dots, x_n) \|_\alpha, \quad \alpha \geq 1,$$

the above maximization problem is
equivalent to

$$K \geq \max_{\substack{\|x\|_2=1 \\ x_i \geq 0}} \sum_{i=1}^n x_i = \max_{\substack{x_i \geq 0 \\ \lambda \neq 0}} \sum_{i=1}^n x_i + \lambda \left(1 - \sum_{i=1}^n (x_i)^2\right) \quad (1c)$$

using Lagrange multipliers.

$$\therefore 0 = \frac{\partial}{\partial x_i} \Rightarrow 1 - 2\lambda x_i = 0 \quad i=1, 2, \dots, n$$

$$\therefore 0 = \frac{\partial}{\partial \lambda} \Rightarrow 1 - \sum_{i=1}^n (x_i)^2 = 0$$

$$\Rightarrow x_i = \frac{1}{2\lambda}, \quad i=1, 2, \dots, n \quad \text{and}$$

$$\text{then, } \lambda = \frac{\sqrt{n}}{2} \quad \& \quad x_i = \frac{1}{\sqrt{n}}$$

$$\therefore \max_{\substack{\|x\|_2=1 \\ x_i \geq 0}} \sum x_i = \frac{n}{\sqrt{n}} = \sqrt{n}$$

$\therefore$ We can take $\boxed{K = \sqrt{n}}$. Moreover,

we now know that this is the smallest possible value for K . $\square$